

Syllabus of ECE253/CSE208: Introduction to Information Theory

- Lecture time/location: TuTh 3:20PM - 4:55PM at E2-192
- Instructor: Dr. Yu Zhang (email: yzhan419@ucsc.edu)
- Instructor office hours: Mondays 3–4pm at E2-547B
- Reader: TBA.
- Course websites: Canvas & Gradescope (for homework and exam submissions)

Course descriptions: An introduction to information theory including topics such as entropy, relative entropy, mutual information, asymptotic equipartition property, channel capacity, differential entropy, and rate distortion theory.

Prerequisites: Calculus; probability and stochastic processes (e.g., Math 3 or AMS 3, 5, 7).

Textbook

R1) Thomas Cover and Joy Thomas, *Elements of Information Theory (2nd Edition)*.

Selected readings

- S1) James Stone, *Information Theory: A Tutorial Introduction*.
 S2) David MacKay, *Information Theory, Inference and Learning Algorithms*
 S3) Yihong Wu and Yury Polyanskiy, *Information Theory: From Coding to Learning*.
 S4) John Pierce, *An Introduction to Information Theory: Symbols, Signals and Noise*.

Tentative agenda

Week	Topic (Source: chapters of textbook R1)
Week-1	Course overview; probability theory revisited
Week-2,3	Entropy, mutual information, and inequalities (chap 2)
Week-4	AEP, entropy rate (chap 3-4)
Week-5,6	Data compression and gambling (chap 5-6)
Week-7	Channel capacity (chap 7);
Week-8	Differential entropy (chap 8); Gaussian channel I (chap 9);
Week-9	Gaussian channel II (chap 9)
Week-10	Rate distortion theory (chap 10);

Key dates:

- In-class Midterm Exam: Thursday, Feb 12, 2026
- Take-home (24-hour) Final Exam: Tuesday, Mar 17, 2026

Score Range	Letter Grade
[93, 100]	A+
[86, 92]	A
[80, 85]	A–
[75, 79]	B+
[70, 74]	B
[65, 69]	C
[60, 64]	D
< 60	Fail

Table 1: Grading scale. A final grade of B or above is considered satisfactory.

Grading: HW 30% + Midterm 40% + Final exam 30%

Academic integrity: Each student must understand the university policy on academic integrity, as well as each instructor’s policy on acceptable collaboration and cheating. If you are unsure whether a particular behavior violates university or faculty policy, it is your responsibility to consult with the instructor for clarification. If you are having trouble in a class, access the success resources available to you – it is far better to seek help than to resort to dishonest academic behaviors. Complete information about academic integrity policy, procedure, and resources is available at http://www.ue.ucsc.edu/academic_misconduct.

Supporting services: UC Santa Cruz is committed to creating an academic environment that supports its diverse student body. If you are a student with a disability who requires accommodations to achieve equal access in this course, please contact the Disability Resource Center (DRC) for the application and relevant paperwork.